

WASP-52b

R-0439

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_171109 · created 2026-07-19T13:51:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458066.6482 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.026
Transit depth (Rp/Rs)^2	3.74 +/- 0.99 [%]
Orbital Inclination (inc)	84.87 +/- 1.55
Airmass coefficient 1 (a1)	1.011 +/- 0.024
Airmass coefficient 2 (a2)	-0.013 +/- 0.067
Scatter in the residuals of the lightcurve fit is	2.39 %
Best Comparison Star	#3 - [437, 211]
Optimal Aperture	3.76
Optimal Annulus	8.85
Transit Duration (day)	0.073 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.026 vs 0.1646 (deviation 1.09σ)
Duration fit vs published	0.073 d vs 0.0758 d (deviation 0.24σ)
Mid-time O–C	1.1 min
Depth SNR	7.4
Residual scatter	2.39 %

Light curve

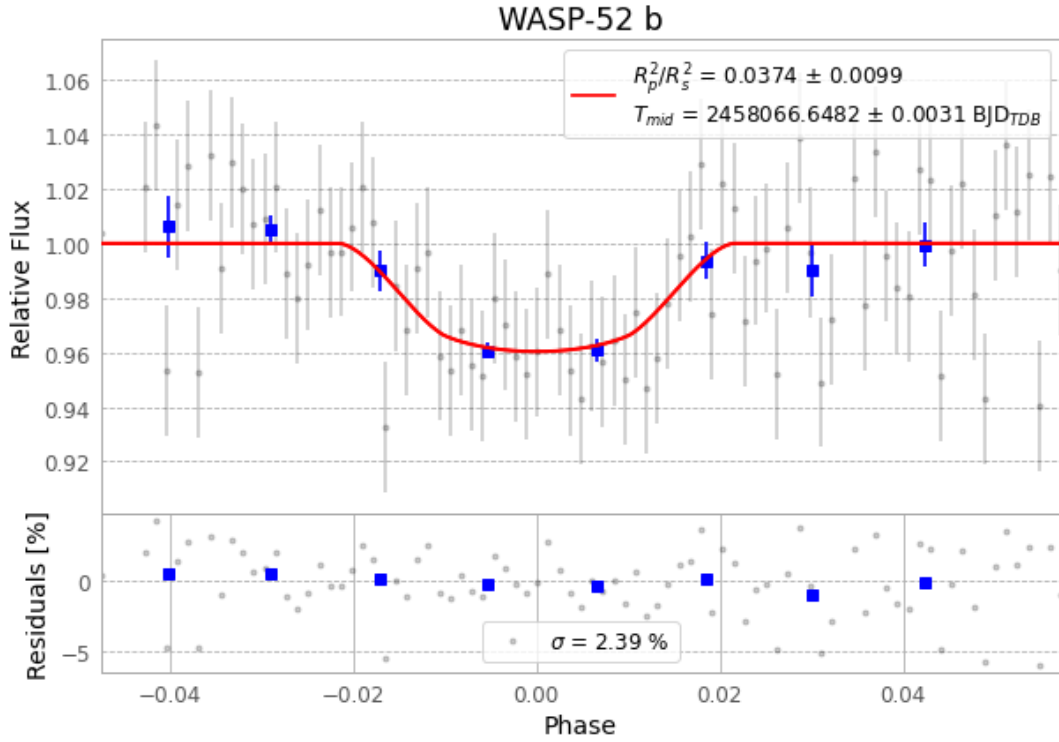


Figure 1 — FinalLightCurve_WASP-52 b_2017-11-08.png (SHA-256 6bba934ece02b5bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9c7678ce6eb02105...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

89 FITS frames (58 MB), WASP-52171109012712.FITS → WASP-52171109055112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171109.log (e8da176cf651...), FinalLightCurve_WASP-52 b_2017-11-08.png (6bba934ece02...), FinalLightCurve_WASP-52 b_2017-11-08.csv (b0fbf1a7264d...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 13:51Z.